

AROMA

AI-powered Repository & Object Model Analyzer

IBM Bob IDE Development

Bob Shell sessions that built AROMA

IBM Bob Hackathon 2026 - Turn idea into impact faster
Built with IBM Bob IDE | Powered by IBM watsonx.ai Granite

IBM Bob as Development Partner

IBM Bob served as the primary AI development partner throughout the AROMA build. Bob Shell was used via CLI for all five development sessions, providing code review, architecture analysis, and verification of the IBM watsonx.ai integration.

Session Summary

#	Session	Focus	Bob Mode
1	Project Init	Full project review	Ask + Code review
2	watsonx Integration	5 API routes, Granite	Code review
3	Security Scanner	OWASP Top 10, CWE	Security audit
4	Smart Refactor	Pattern detection	Refactoring
5	Final Review	Full audit	Review + cleanup

Session Reports in Repository

All five Bob Shell task session reports are stored in `bob_sessions/` directory. Each folder contains:

- `task-history.md`: Full exported task history from Bob Shell
- `screenshot.png`: Task session consumption summary

Session Details

Session 1: Project Initialization

Bob reviewed the entire AROMA project, identified all 6 modules, and confirmed the IBM watsonx.ai integration architecture. Verified Granite model config and IAM token management.

Session 2: watsonx.ai Integration

Bob reviewed all 5 API routes and verified watsonx.ai integration. Confirmed temperature settings, JSON extraction, and fallback strategies for each route.

Session 3: Security Scanner

Bob reviewed the Security Scanner component and API route, verifying OWASP Top 10 detection with CWE references. Confirmed security score calculation.

Session 4: Smart Refactor

Bob reviewed the Smart Refactor module, suggested improvements to pattern detection. Verified severity classification and modern alternatives.

Session 5: Final Code Review

Bob performed comprehensive final review, checking for non-IBM AI references. Verified all `.env` credentials gitignored, no API keys exposed.

How AROMA Embodies the Hackathon Theme

IBM Bob Capability Used	AROMA Module
Explained existing code logic	Code Explorer
Traced data flows across files	Flow Tracer
Detected anti-patterns	Smart Refactor
Generated documentation	Doc & Test Generator
Identified security issues	Security Scanner